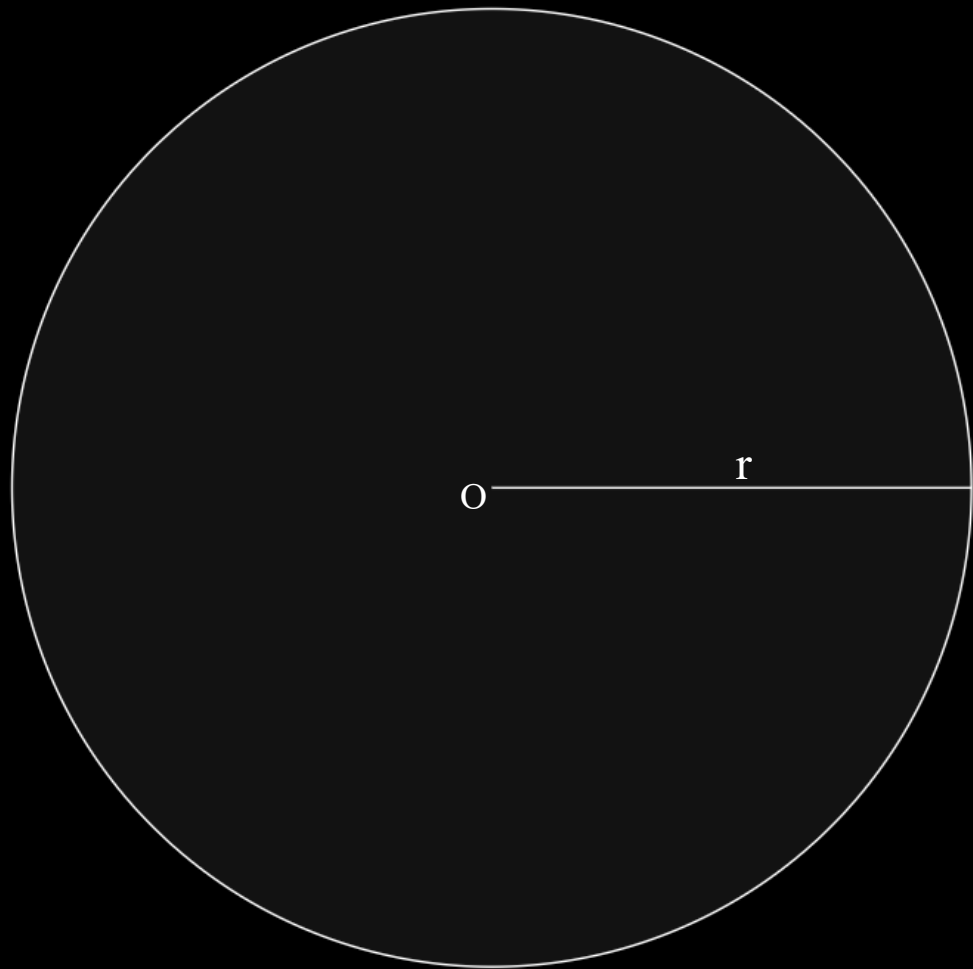


Finding π value

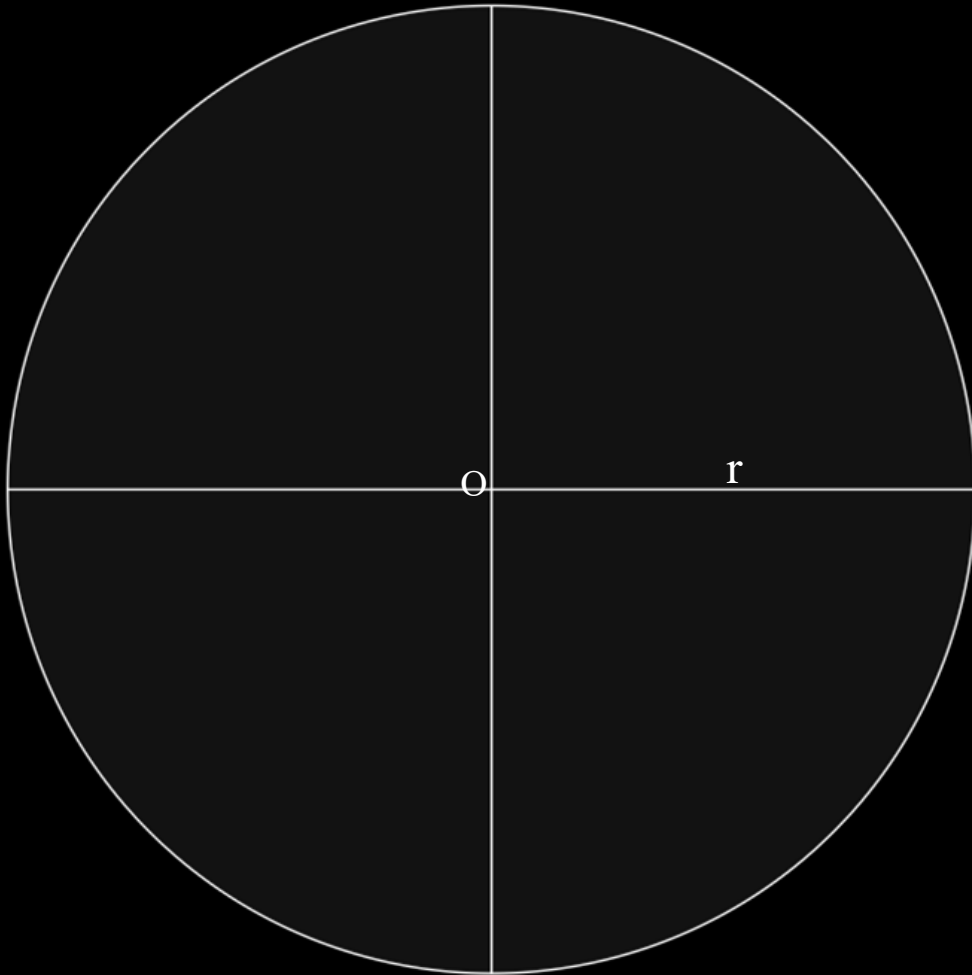
Basic Geometry · Pythagorean Theorem · Pattern Recognition

Independently Rediscovered by Adithya M

[Derived independently; later discovered this connects to Archimedes' method, François Viète's formula and Nested radicals of cosine]

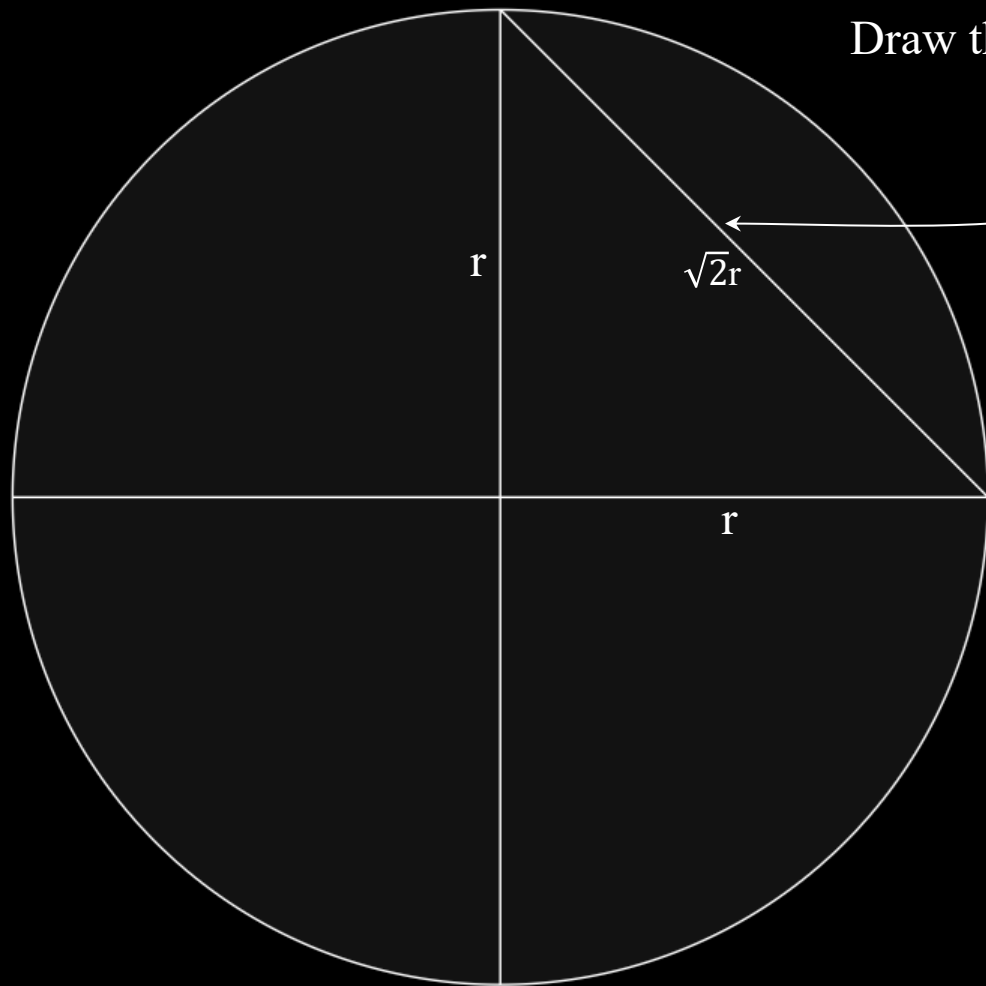


Consider a CIRCLE
of radius 'r' with
centre O



Divide the circle into 4 equal quadrants.

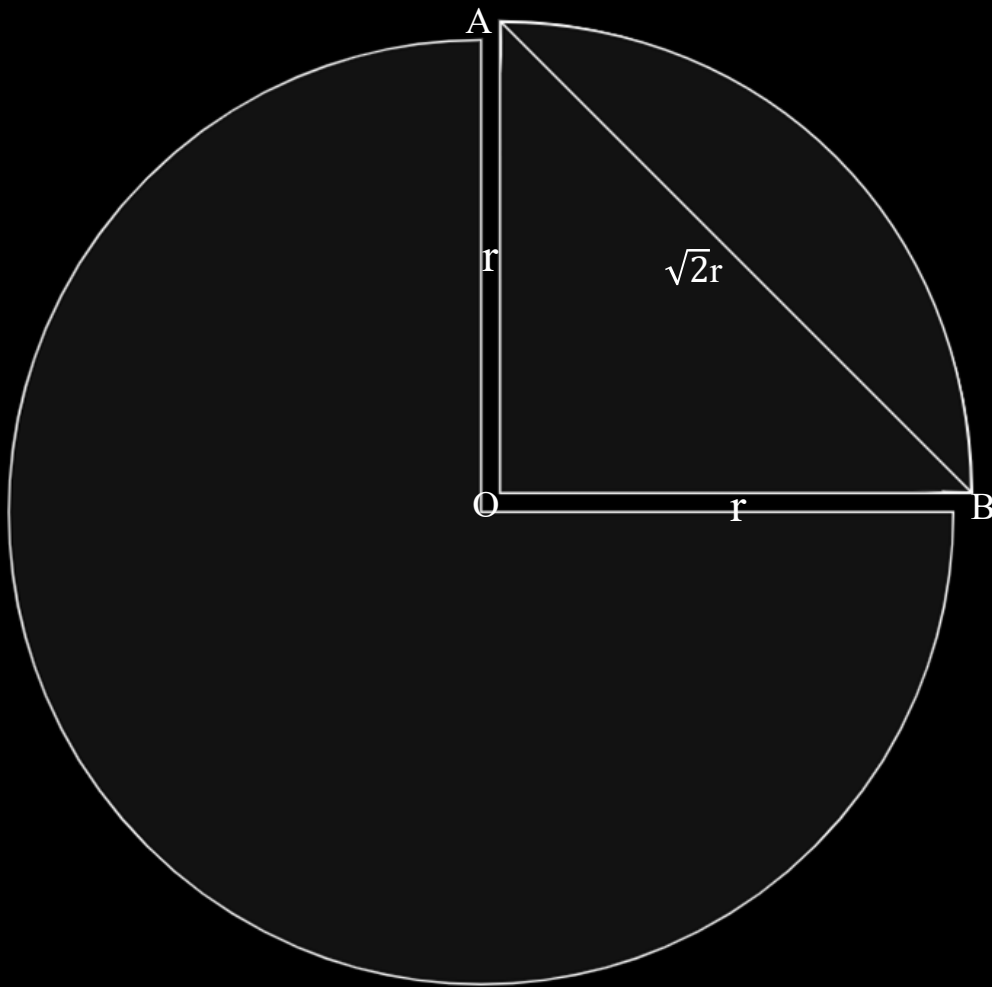
$$\text{Area of Circle} = 4 \times \text{Area(quarter of circle)}$$



Draw the hypotenuse line making a right triangle

Using Pythagorean theorem,
Hypotenuse length = $\sqrt{2}r$

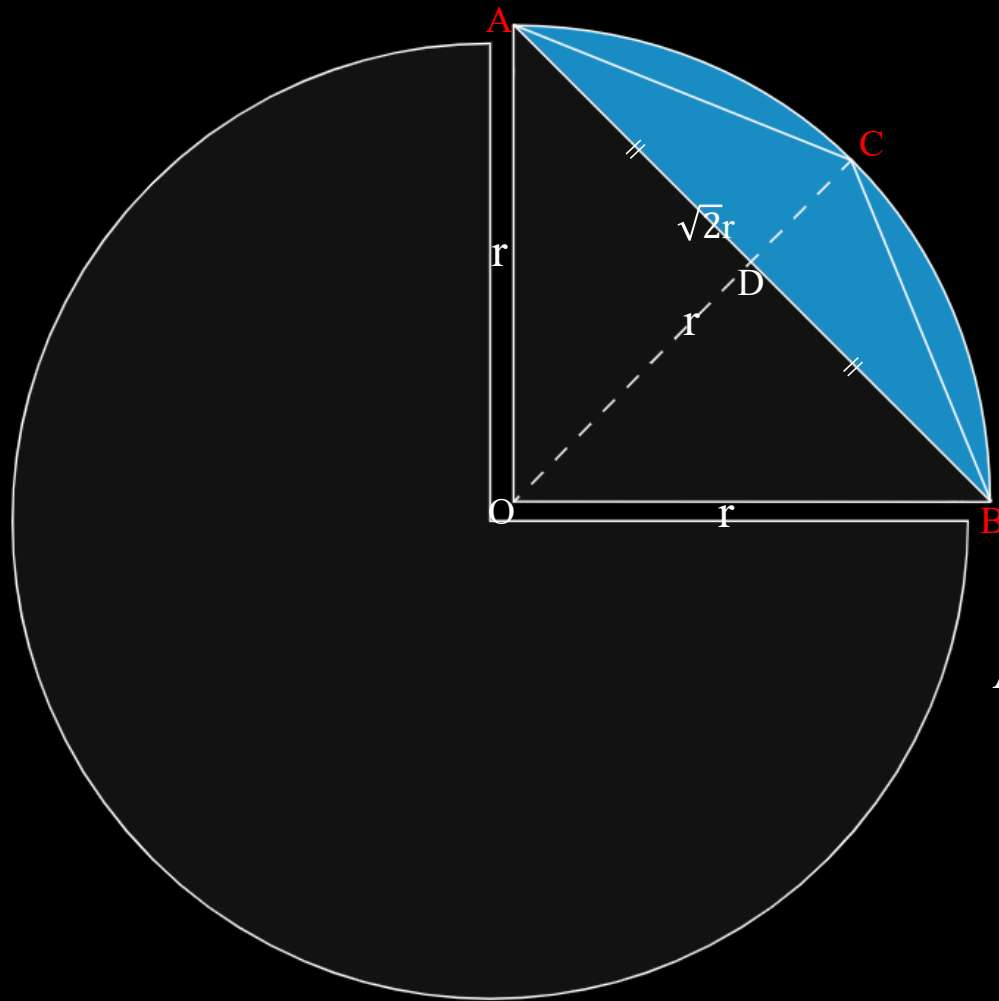
$$\text{Area of triangle} = \frac{bh}{2} = \frac{r^2}{2}$$



$$\text{Area}(\text{quadrant}) = \text{Area}(\text{triangle}) + \text{Area}(\text{segment})$$

$$\text{Area}(\text{triangle}) = \frac{r^2}{2}$$

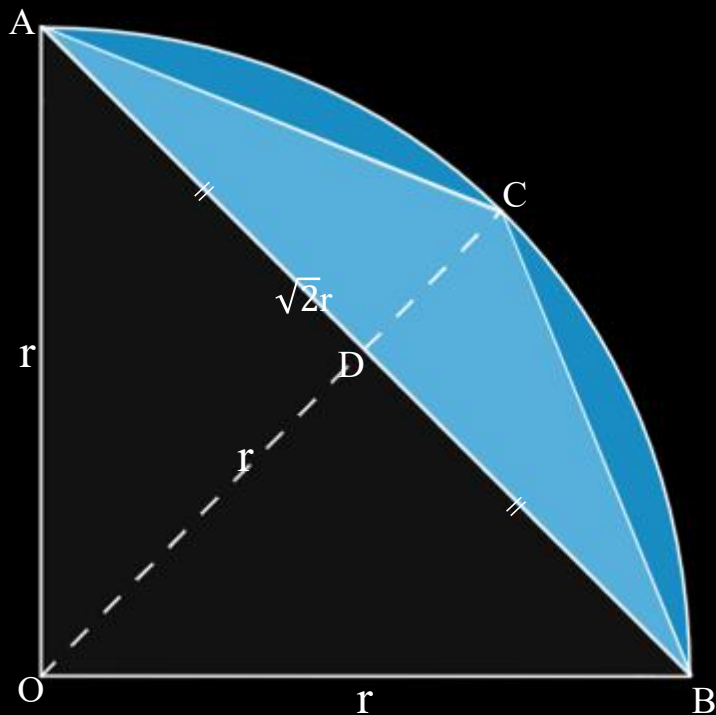
$$\text{Area}(\text{segment}) = ?$$



Draw a line OC from centre O to edge of circle which perpendicularly bisects the chord into 2 equal lengths.

Join AC and BC.

$$\text{Area}(\text{segment ABC}) = \text{Area}(\text{triangle ABC}) + 2 \times \text{Area}(\text{segment AC})$$



$$\text{Area}(\text{triangle } ABC) = 2 \times \text{Area}(\text{triangle } ADC)$$

$$\text{Area}(\text{triangle } ADC) = \frac{(AD)(DC)}{2}$$

$$AD = \frac{AB}{2} = \frac{\sqrt{2}r}{2}$$

$$DC = OC - OD$$

$$= r - \sqrt{r^2 - \left(\frac{\sqrt{2}r}{2}\right)^2}$$

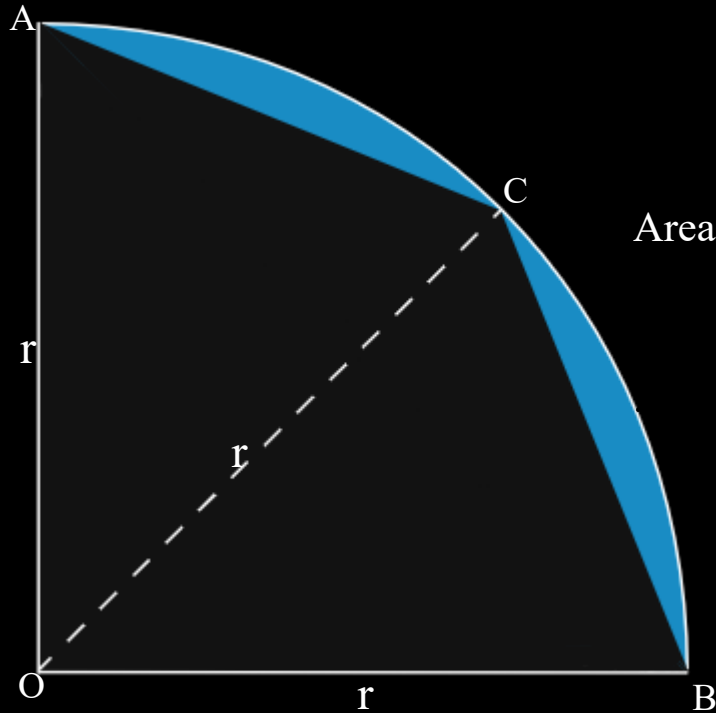
$$= \frac{r(2-\sqrt{2})}{2}$$

$$\text{Area}(\text{triangle } ADC) = \frac{r^2(\sqrt{2}-1)}{4}$$

$$\text{Area}(\text{triangle } ABC) = \frac{r^2(\sqrt{2}-1)}{2}$$

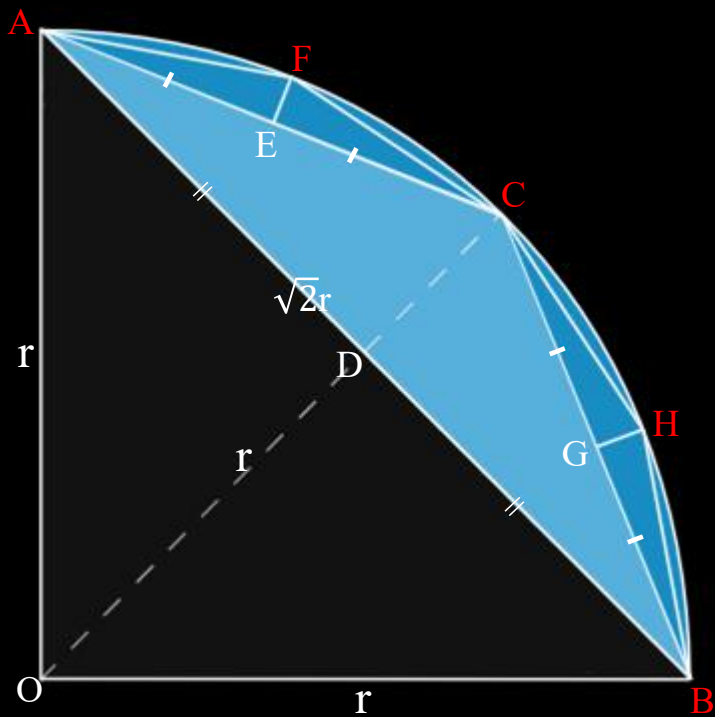
$$\text{Area}(\text{segment } AC) = ?$$

Summarising what we did:



$$\begin{aligned}\text{Area of Circle} &= 4 \times \text{Area}(\text{quarter of circle}) \\ &= 4 \times [\text{Area}(\text{triangle AOB}) + \text{Area}(\text{segment ABC})] \\ &= 4 \times \left[\frac{r^2}{2} + \text{Area}(\text{triangle ABC}) + 2 \times \text{Area}(\text{segment AC}) \right] \\ &= 4 \times \left[\frac{r^2}{2} + \frac{r^2(\sqrt{2}-1)}{2} + 2 \times \text{Area}(\text{segment AC}) \right]\end{aligned}$$

$$\text{Area}(\text{segment AC}) = ?$$

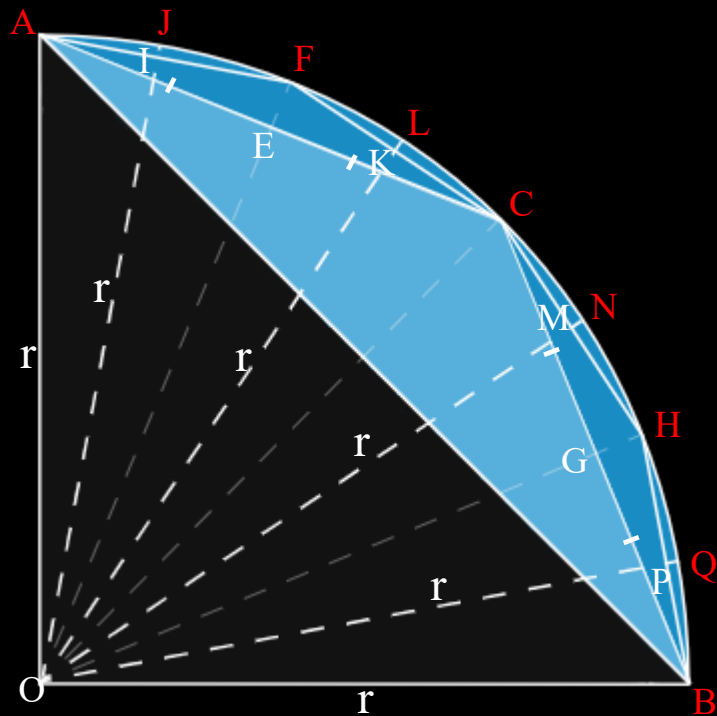


Draw lines EF and GH which perpendicularly bisects the chords AC and BC into 2 equal lengths respectively.

Join **AF**, **CF**, **CH** and **BH**.

$$\text{Area}(\text{segment AC}) = \text{Area}(\text{triangle ACF}) + 2 \times \text{Area}(\text{segment AF})$$

(or segment AFC)

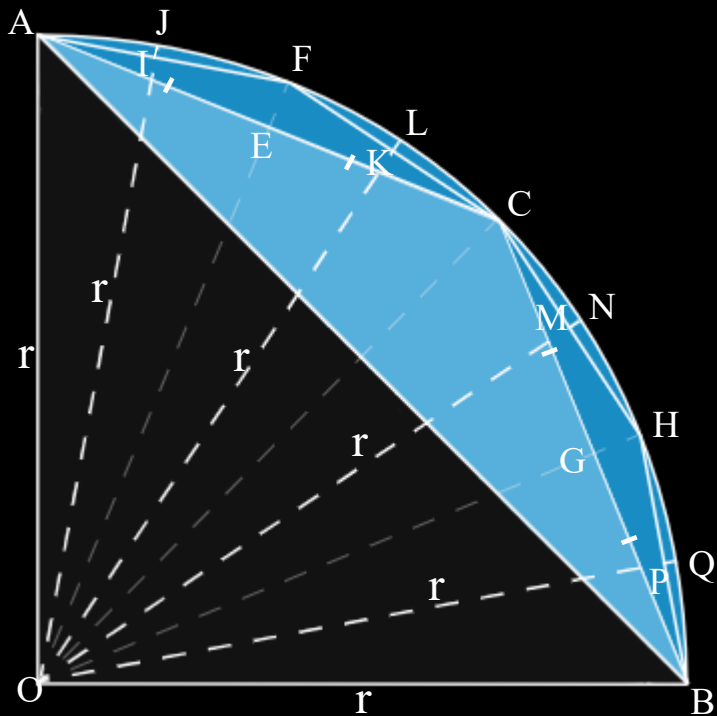


Similar to earlier, draw lines IJ, KL, MN, PQ which perpendicularly bisects the chords AF, FC, CH, HB into 2 equal lengths respectively.

Join **AJ**, **JF**, **FL**, **LC**, **CN**, **NH**, **HQ** and **QB**.

[Not shown in diagram due to visibility constrain.]

$$\begin{aligned} \text{Area}(\text{segment AF}) &= \text{Area}(\text{triangle AFJ}) + \\ &\quad (\text{or segment AJF}) \qquad 2 \times \text{Area}(\text{segment AJ}) \end{aligned}$$



$$\text{Area}(\text{triangle AFJ}) = 2 \times \text{Area}(\text{triangle AIJ})$$

$$\text{Area}(\text{triangle AIJ}) = \frac{(AI)(IJ)}{2}$$

$$AI = \frac{AF}{2} = \frac{\sqrt{AE^2 + EF^2}}{2} = \frac{\sqrt{\left(\frac{r\sqrt{2-\sqrt{2}}}{2}\right)^2 + \left(\frac{r(2-\sqrt{2+\sqrt{2}})}{2}\right)^2}}{2} = \frac{r\sqrt{2-\sqrt{2+\sqrt{2}}}}{2}$$

$$IJ = OJ - OI$$

$$= r - \sqrt{r^2 - \left(\frac{r\sqrt{2-\sqrt{2+\sqrt{2}}}}{2}\right)^2}$$

$$= \frac{r(2 - \sqrt{2 + \sqrt{2 + \sqrt{2}}})}{2}$$

$$\text{Area}(\text{triangle AIJ}) = \frac{r^2(2\sqrt{2-\sqrt{2+\sqrt{2}}} - \sqrt{2-\sqrt{2}})}{8}$$

$$\text{Area}(\text{triangle AFJ}) = \frac{r^2(2\sqrt{2-\sqrt{2+\sqrt{2}}} - \sqrt{2-\sqrt{2}})}{4}$$

$$\text{Area}(\text{segment AJ}) = ?$$

So, now you know where we are going!!

We follow the exact same steps of,

- (1) Drawing perpendicular bisecting lines that divides the chords into 2 equal lengths.
- (2) Forming 2 triangles in each segment by joining chord vertexes with intersecting point of circle and perpendicular bisector.
- (3) Finding area of the triangle formed.
- (4) Finding the area of newly formed segment (from step (2)) by repeating from step (1), indirectly drawing **2n-polygons inscribed in a circle** and finding area of it.

Summarising what we did:

$$\begin{aligned}
 \text{Area of Circle} &= 4 \times \left[\frac{r^2}{2} + \frac{r^2(\sqrt{2}-1)}{2} + 2 \times \text{Area}(\text{segment AC}) \right] \\
 &= 4 \times \left[\frac{r^2}{2} + \frac{r^2(\sqrt{2}-1)}{2} + 2[\text{Area}(\text{triangle ACF}) + 2 \times \text{Area}(\text{segment AF})] \right] \\
 &= 4 \times \left[\frac{r^2}{2} + \frac{r^2(\sqrt{2}-1)}{2} + 2\left[\frac{r^2(2\sqrt{2}-\sqrt{2}-\sqrt{2})}{4} + 2[\text{Area}(\text{triangle AFJ}) + 2 \times \text{Area}(\text{segment AJ})]\right] \right] \\
 &= 4 \times \left[\frac{r^2}{2} + \frac{r^2(\sqrt{2}-1)}{2} + 2\left[\frac{r^2(2\sqrt{2}-\sqrt{2}-\sqrt{2})}{4} + 2\left[\frac{r^2(2\sqrt{2-\sqrt{2}+\sqrt{2}}-\sqrt{2-\sqrt{2}})}{4} + 2 \times \text{Area}(\text{segment AJ})\right] \right] \right] \\
 &= 4 \times \left[\frac{r^2}{2} + \frac{r^2(\sqrt{2}-1)}{2} + 2\left[\frac{r^2(2\sqrt{2}-\sqrt{2}-\sqrt{2})}{4} + 2\left[\frac{r^2(2\sqrt{2-\sqrt{2}+\sqrt{2}}-\sqrt{2-\sqrt{2}})}{4} + 2 \times [\dots + 2 \times [\dots]] \right] \right] \right] \\
 &\hspace{15em} [This continues forever...]
 \end{aligned}$$

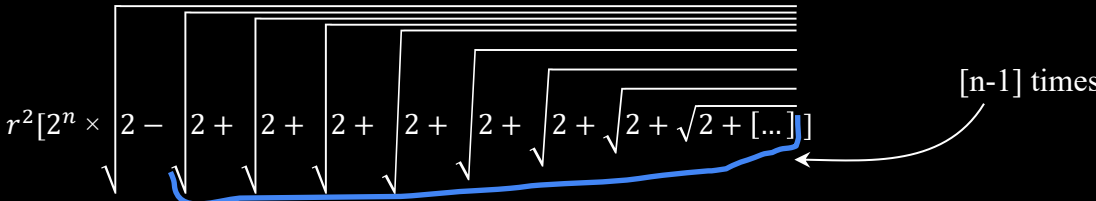
Though there are infinite number of terms, we get to observe a clear pattern in above expression!

The Pattern:

Total no. of outermost chords: (Polygon sides with radius r)	Area enclosed by polygon:	On Simplification (1):
4	$4 \times \left[\frac{r^2}{2}\right]$	$r^2 [2]$
8	$4 \times \left[\frac{r^2}{2} + \frac{r^2(\sqrt{2}-1)}{2}\right]$	$r^2 [2 + 2(\sqrt{2} - 1)]$
16	$4 \times \left[\frac{r^2}{2} + \frac{r^2(\sqrt{2}-1)}{2} + 2\left[\frac{r^2(2\sqrt{2}-\sqrt{2}-\sqrt{2})}{4}\right]\right]$	$r^2[2 + 2(\sqrt{2} - 1) + 2(2\sqrt{2} - \sqrt{2} - \sqrt{2})]$
32	$4 \times \left[\frac{r^2}{2} + \frac{r^2(\sqrt{2}-1)}{2} + 2\left[\frac{r^2(2\sqrt{2}-\sqrt{2}-\sqrt{2})}{4} + 2\left[\frac{r^2(2\sqrt{2-\sqrt{2}+\sqrt{2}}-\sqrt{2-\sqrt{2}})}{4}\right]\right]\right]$	$r^2[2 + 2(\sqrt{2} - 1) + 2(2\sqrt{2} - \sqrt{2} - \sqrt{2}) + 4(2\sqrt{2-\sqrt{2}+\sqrt{2}} - \sqrt{2-\sqrt{2}})]$
64*	$4 \times \left[\frac{r^2}{2} + \frac{r^2(\sqrt{2}-1)}{2} + 2\left[\frac{r^2(2\sqrt{2}-\sqrt{2}-\sqrt{2})}{4} + 2\left[\frac{r^2(2\sqrt{2-\sqrt{2}+\sqrt{2}}-\sqrt{2-\sqrt{2}})}{4} + 2\left[\frac{r^2(2\sqrt{2-\sqrt{2}+\sqrt{2}}-\sqrt{2-\sqrt{2}+\sqrt{2}})}{4}\right]\right]\right]\right]$	$r^2[2 + 2(\sqrt{2} - 1) + 2(2\sqrt{2} - \sqrt{2} - \sqrt{2}) + 2(2\sqrt{2-\sqrt{2}+\sqrt{2}} - \sqrt{2-\sqrt{2}}) + 8(2\sqrt{2-\sqrt{2}+\sqrt{2}} - \sqrt{2-\sqrt{2}+\sqrt{2}})]$

**Can be manually verified by geometry and calculation.*

The Pattern:

Total no. of outermost chords: (Polygon sides with radius r)	On Simplification (1):	On Simplification (2):
4	$r^2[2]$	$r^2[2]$
8	$r^2 [2 + 2(\sqrt{2} - 1)]$	$r^2[2^1 \times \sqrt{2}]$
16	$r^2[2 + 2(\sqrt{2} - 1) + 2(2\sqrt{2 - \sqrt{2}} - \sqrt{2})]$	$r^2[2^2 \times \sqrt{2 - \sqrt{2}}]$
32	$r^2[2 + 2(\sqrt{2} - 1) + 2(2\sqrt{2 - \sqrt{2}} - \sqrt{2}) + 4(2\sqrt{2 - \sqrt{2 + \sqrt{2}}} - \sqrt{2 - \sqrt{2}})]$	$r^2[2^3 \times \sqrt{2 - \sqrt{2 + \sqrt{2}}}]$
64	$r^2[2 + 2(\sqrt{2} - 1) + 2(2\sqrt{2 - \sqrt{2}} - \sqrt{2}) + 2(2\sqrt{2 - \sqrt{2 + \sqrt{2}}} - \sqrt{2 - \sqrt{2}}) + 8(2\sqrt{2 - \sqrt{2 + \sqrt{2 + \sqrt{2}}}} - \sqrt{2 - \sqrt{2 + \sqrt{2}}})]$	$r^2[2^4 \times \sqrt{2 - \sqrt{2 + \sqrt{2 + \sqrt{2}}}}]$
n	 $r^2[2^n \times \sqrt{2 - \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}}}}}}]$ [n-1] times	

Area of polygon with 'n' sides = $r^2 [2^n \times \sqrt{2 - \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}}}}}]$

of radius 'r'

[n-1] times

Area of Circle of radius 'r' = $r^2 \times \lim_{n \rightarrow \infty} 2^n \sqrt{2 - \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}}$

($n \rightarrow \infty$)

[As $n \rightarrow \infty$, the inscribed polygon converges to the circle, and hence its area converges to πr^2]

n square roots

Area of Circle of radius 1 = $\lim_{n \rightarrow \infty} 2^n \sqrt{2 - \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}}$

[So that r^2 is removed, yielding a constant value]

n square roots

$$\text{Value of } \pi = \lim_{k \rightarrow \infty} 2^k \sqrt{2 - \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}}$$

k square roots

$$\text{Value of } \pi = \lim_{k \rightarrow \infty} 2^k \sqrt[k]{2 - \sqrt{2 + \sqrt{2 + \sqrt{2 + \dots}}}}$$

k square roots

'k' value (Iteration count)	$\approx \pi$ value:
1	2.828427124746190
2	3.061467458920718
3	3.121445152258052
4	3.136548490545939
5	3.140331156954753
10	3.141591421511200
20	3.141592653588618
50	3.1415926535897932384626433832784

Fact:

The convergence rate of above formula is ≈ 0.602 decimal digits per iteration. Example,

10 iterations $\rightarrow \approx 6$ correct digits of π

50 iterations $\rightarrow \approx 30$ correct digits of π

167 iterations $\rightarrow \approx 100$ correct digits of π

Fun Fact:

In real life practical use, we require,

6 digits $\rightarrow$ Measure Earth to meter precision

15 digits $\rightarrow$ Run space missions

40 digits $\rightarrow$ Measure the universe smaller than a proton

“Mathematics hides infinity inside simple shapes.”
A geometric exploration driven by curiosity.

Always learning — open to conversations.

LinkedIn: www.linkedin.com/in/adithya3907/

X (Twitter): <https://x.com/adithya3907>